

Quantitative criteria for native energetic heterogeneity influences in the prediction of protein folding kinetics

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Energy landscape theory requires that the protein-folding mechanism is generally globally directed or funneled toward the native state. The collective nature of transition state ensembles further suggests that sufficient averaging of the native interactions can occur so that the knowledge of the native topology may suffice for predicting the mechanism. Nevertheless, while simple homogeneously weighted native topology-based models predict the folding mechanisms for many proteins, for other proteins knowledge of the native topology, by itself, seems not to suffice in determining the folding mechanism. Simulations of proteins with differing topologies reveal that the failure of homogeneously weighted topology-based models can, however, be completely understood within the framework of a funneled energy landscape and can be quantified by comparing the fluctuation of entropy cost for forming contacts to the expected fluctuations in contact energy. To be precise, we find the transition state ensembles of proteins with all- α topologies, which are more uniform in the specific entropy cost of contact formation, have transition state ensembles that are more readily perturbed by differences in energetic weights than are the transition state ensembles of proteins with significant amounts of β -structure, where the specific entropy costs of contact formation are more widely distributed. This behavior is consistent with a random-field Ising model analogy that follows from the free energy functional approach to folding.

energy landscape theory | random-field Ising model | native topology-based models

Evolution has sculpted the energy landscapes of natural proteins to be globally directed toward the native state (1). More precisely, sequences have evolved so that the interactions present in the native state are more stabilizing than extreme value statistics would lead one to expect for forming random contact interactions (2). In this case, nonnative interactions provide primarily a frictional influence (3, 4). If, furthermore, the nonnative interactions were so weak that they can be completely neglected in comparison to those in the native structure, the balance between the chain entropy and native interaction energies alone would determine the folding mechanism. Reflecting this balance, even crude topology-based measures, such as contact order, provide rough estimates of the folding rates of two-state folding proteins (5). Because many contacts are formed in the transition state ensemble, it further becomes reasonable to simplify the model by replacing individual contact energies with an average value, neglecting sequence variability. The resulting energy landscape is perfectly funneled, but now encodes only the native topology (6). Such averaged contact energy models predict the folding rate of proteins in many cases (7), even when the simple contact order estimate is not very accurate (8). Many studies have shown that a wide range of details of folding and binding mechanisms, such as whether specific intermediates form or not, are also correctly predicted by such native topology-based models in many cases (6, 9). In some circumstances where seemingly minor differences of topology are involved, even predicting mechanistic subtleties is possible from this homogeneous model (10). More quantitative features about the structure of the transition state ensembles, such as the Φ values, are also generally

well predicted by pure topology models (11–13), but at this level more discrepancies appear (13). These discrepancies caution us that while the successes of pure native topology-based models are impressive, one must examine the homogeneity assumption that is made in topology-based modeling, which averages the native contact energies. In quantitative terms, can we determine when the homogeneity assumption will suffice and when it will not?

Failures of the contact averaging approximation were first noted in studying structurally homologous proteins having disparate sequences but essentially the same topology. According to the averaging ansatz, even if such proteins are distantly related in sequence, they should exhibit similar folding mechanisms because they share the same native contact pattern. A striking example of the seeming validity of the averaging approximation occurs in the folding of the src and spectrin SH3 domains, which both have the same all- β topology. Even though they have low sequence homology (27%), they are experimentally observed to exhibit very similar transition state ensembles, and this behavior is also seen in simulations (14, 15). The structure of the transition state ensemble is also robust to changes in environmental conditions for these systems (14). Another example is provided by comparing the folding of acylphosphatase with the folding of human procaryoxypeptidase A2 activation domain. These proteins both have similar α/β topologies and folding mechanisms while sharing only 13% sequence identity (16), again indicating that the native topology suffices to determine the folding mechanism. Other sets of proteins with nearly identical α/β topologies and low sequence similarity, however, do sometimes exhibit different folding mechanisms, but this often involves symmetry-breaking between two essentially isomorphic folding routes (17–19). The small differences of free energy between two possible routes can easily be determined by just a few contacts. The most dramatic differences in the folding mechanism for topologically equivalent proteins are seen in sets of all- α structural homologues. For Im7 and Im9, both nearly identical 4-helix bundles, the folding mechanism of Im7 involves a populated intermediate, whereas Im9 folds by a 2-state manner, even though there is 60% sequence identity between the two proteins (20). Interestingly, the main transition states still have similar Φ values (21). Recently, Clarke and co-workers (22) showed that the folding rates of α -spectrin repeats of similar topology can vary over several orders of magnitude. Although the native topology clearly plays a critical role in the protein-folding mechanism, these examples imply that energetic weights of the specific residue interactions can sometimes be important as well.

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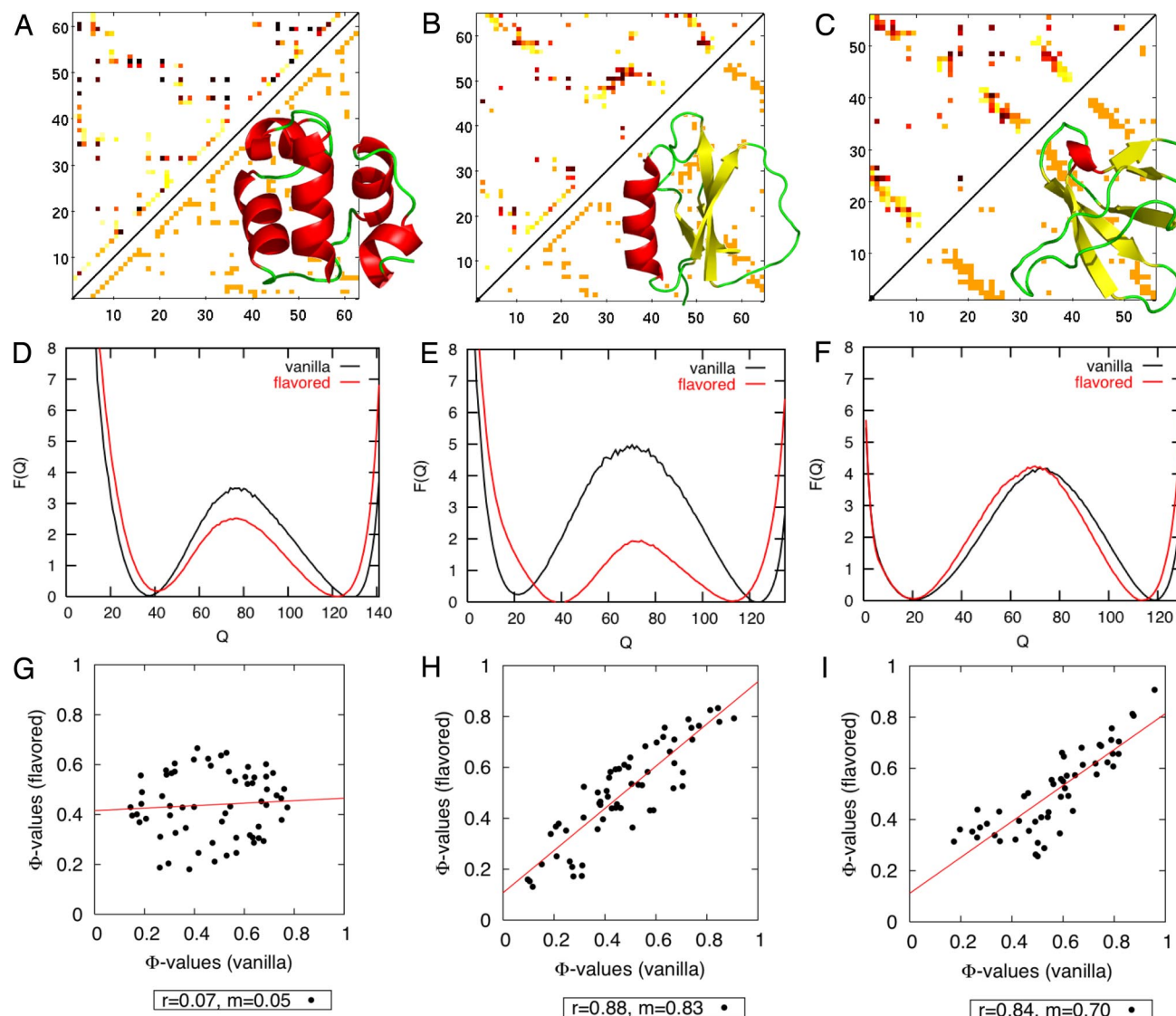


Fig. 1. The folding mechanisms of the all- α lambda repressor [Protein Data Bank (PDB) ID code 1R69] (A, D, and G), the α/β CI2 (PDB ID code 2CI2) (B, E, and H), and all- β src-SH3 domain (PDB ID code 15RL) (C, F, and I). (A–C) The matrices of the interaction energies in the vanilla and flavored native-topology-based models are plotted below and above the diagonal, respectively, with darker colors representing stronger interactions. The corresponding native structures are also shown. (D–F) From simulations of the vanilla and flavored models, the free energy profiles were generated with respect to the order parameter Q . (G–I) The Φ values from the vanilla and flavored models are compared in a plot with a best-fit line; m is the slope of the line.

The effects of energetic heterogeneity of the native interactions on the folding mechanism have already been addressed by using analytical energy landscape theory. Using the free energy functional approach (23, 24), Plotkin and co-workers found that introducing energetic heterogeneity to native interactions in a minimally frustrated system lowers the free energy barrier until it vanishes with a sufficiently large dispersion of native contact energies, and similar behaviors were seen in simulations on lattices (25–27). From the free energy functional perspective, the effects of contact heterogeneity are very much analogous to the well-known phase transition in the random field ferromagnet when the dispersion of site energies become large (28). Sometimes, with sufficiently large dispersion of the native contact energies, the Φ values becomes bimodal, with extreme values close to 0 or 1 (26, 27). Recently, in the context of the α/β CI2 and the all- β src-SH3 domain, Suzuki and Onuchic (29) have shown that the structure of the transition state ensemble is robust and insensitive to energetic details.

In principle, we will directly compare the analytical results of free energy functional approaches with those of native topology-based

model simulations. We have carried out simulations that show that, in keeping with the expectations from analytical theory, homogeneously weighted native topology-based models (based on the averaging approximation) determine the folding mechanism of proteins when the entropy costs of contact formation are widely distributed, but that such models fail when the native contact heterogeneity is sufficiently large, even for the α/β and all- β -protein. For the latter, however, the necessary heterogeneity for the breakdown of the averaging ansatz is larger than seems physically reasonable. This explains why homogeneously weighted native topology-based models with large contact entropy dispersion readily reproduce the folding mechanisms of some proteins, whereas the folding mechanisms of proteins with too narrow a distribution of contact entropies cannot be so easily predicted.

Results and Discussion

Homogeneous Versus Heterogeneous Contact Energies in Funneled Landscapes We begin by comparing simulations of the simple homogeneously weighted C_α models to corresponding simulations

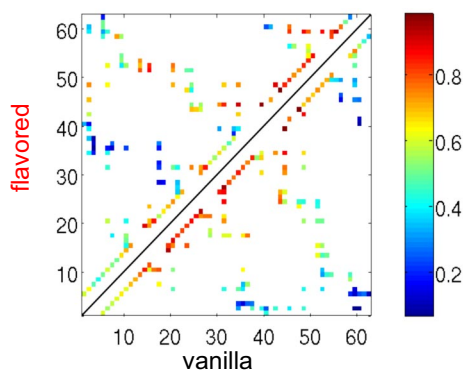


Fig. 2. The probability of a contact in the transition state of the lambda repressor, an all- α protein, with the vanilla and flavored models.

having energetic heterogeneity based on the 20-letter Miyazawa–Jernigan (MJ) contact potential (30). Although the degree of heterogeneity of the MJ potential may be too large, its native contact dispersion is similar to what is predicted by other more refined contact potentials (31). For linguistic simplicity, we will refer to these two variants, both describing perfectly funneled landscapes, as “vanilla” and “flavored” models, respectively. As a starting point, we surveyed several 2-state folding proteins that have been studied previously by both simulations and in the laboratory. We chose the all- α lambda repressor, the α/β CI2, and the all- β src-SH3 domain. In all three proteins, the contact energies in the flavored models seem evenly distributed, with no immediately obvious clusters of either high or low energetic weights (Fig. 1*A–C*). To quantitatively characterize the folding mechanism, we performed the weighted histogram analysis method (WHAM) to calculate thermodynamic quantities with respect to the order parameter, Q , the fraction of native contacts. We recently showed that Q is one of several simple structural reaction coordinates that captures the folding mechanism on smooth landscapes, even for complicated folding mechanisms (32). In the case of the lambda repressor and CI2, a decrease in the free energy barrier is observed (Fig. 1*D* and *E*), as predicted analytically (25, 26). We also note that the unfolded basin free energy minimum occurs at a higher Q (the fraction of native contacts) in the flavored model than in the vanilla model, whereas conversely the folded basin has lower Q . For src-SH3, however, the free energy barrier does not change when the energetic heterogeneity is introduced (Fig. 1*F*). For both CI2 and src-SH3, the Φ values derived from the simulations of vanilla and flavored models are very similar, with correlation coefficients greater than 0.70 (Fig. 1*H* and *I*). In contrast, the Φ values for the lambda repressor predicted by the vanilla and fully flavored models are essentially uncorrelated with each other (Fig. 1*G*). A closer analysis of the transition state ensemble for the vanilla model reveals that the folding nucleus consists of structured second and third helices with largely unformed long-range interactions (Fig. 2). In contrast, the transition state ensemble of the flavored model predominantly includes structured long-range interactions between the second and fourth helices (Fig. 2). Oas and co-workers (33) performed NMR spectroscopy of 7 alanine-to-glycine mutants of the lambda repressor, and their limited observations indicated that the first and fourth helices are most populated in the transition state ensemble, whereas the second, third, and fifth helices are less populated. It seems that, although not precisely reproducing the experiment, the flavored model agrees more with the pattern of experimental results than does the vanilla model. To determine whether the short-range interaction energies are the source of the discrepancy between the folding mechanisms observed in the vanilla and flavored models, an inhomogeneous model was also simulated where only the contact energies of the short-range

interaction energies of the flavored model were changed back to those of the vanilla model. Now, the free energy barrier becomes about the same as that for the vanilla model [supporting information (SI) Fig. S1*A*], but one still finds the poor correlation between the Φ values in this partially flavored model and the homogeneous case (Fig. S1*B*).

We also simulated several other representative all- α protein domains that we selected from the CATH database (34) (CATH IDs: 1v54E0, 1f6vA0, and 1cy5A0). We chose these proteins because they capture a diverse range in the degree of short-range vs. long-range interactions, as well as helical content (Fig. 3*A–C*). The contact map of 1v54E0 contains mostly relatively short-range interactions (Fig. 3*A*), whereas 1cy5A0 has a large number of long-range interactions (Fig. 3*C*). 1f6vA0 has an intermediate number of long-range interactions (Fig. 3*B*). Again, the energetic weights seem to be evenly distributed across all of the native interactions (Fig. 3*A–C*). In all three cases, the flavored model yields a lower free energy barrier than the vanilla model and the folded basin has a lower Q for the flavored model (Fig. 3*D–F*). For 1f6vA0, the peak of the free energy barrier occurs at a lower Q in the flavored model (Fig. 3*E*). In each case, the Φ values predicted by the vanilla and flavored models for these all- α proteins exhibit no significant correlation (Fig. 3*G–I*).

Energetic and Entropic Fluctuations in the Folding Mechanism. The differences in the topologies of all- α and all- β proteins can be quantified by the ratio of the number of long-range interactions versus short-range interactions ($N_{\text{long}}/N_{\text{short}}$). Three different peaks appear in the distribution of $N_{\text{long}}/N_{\text{short}}$ for the nonredundant set of the PDB, corresponding to the all- α , α/β , and all- β topologies (Fig. S2*A*). These peaks are also observed when only proteins that have been shown to be 2-state folders are included (Fig. S2*B*). All- α proteins have proportionally the lowest number of long-range interactions because the intrahelical interactions stabilize the secondary structure. For all- β proteins, numerous long-range interactions must form between individual sheets.

To examine the interplay between energetic and entropic contributions to folding, the energy and entropy lost upon formation of native contacts is calculated for the lambda repressor, CI2, and src-SH3 domain (Fig. 4). The energy, $E(Q)$, can be readily calculated as a summation of the inhomogeneous energetic weights, ε_{ij} , of the native interactions (ij) for the native contacts made (Q_{ij}): $E(Q) = +\sum_{ij} \varepsilon_{ij} Q_{ij}$. Similarly, the entropy, $S(Q)$, can be represented approximately as a summation of the entropy (S_{ij}) lost upon forming native contacts in the context of an already partially formed ensemble of structures: $S(Q) = +\sum_{ij} S_{ij} Q_{ij}$. A reasonable approximation to S_{ij} can be found by following an approach similar to that of Shoemaker *et al.* (24). They suggested that initially the entropy lost in sequentially forming short-range interactions can be approximated by the Jacobson–Stockmayer formula (35), $S_{ij} = +k_B \log[\Delta V/|i - j|^{3/2}]$.

Assuming that the denatured protein can be modeled as a random flight chain, the quantity $\Delta V = [(3/2)\pi]^{3/2} \Delta \tau / l_0^3$, where $\Delta \tau$ is the volume of the interaction range and l_0 is the persistence length. But Shoemaker *et al.* (24) also argued that if some structure is already formed, any entropy lost will continue to make sequentially distant interactions and saturate to that of a typical fluctuating segment of a chain, as introduced by Flory in the mean field theory of rubber vulcanization. This yields $S_{ij} = +k_B \log[\Delta V/(\mu/N)^{3/2}]$ where μ is the number of contacts made and N is the number of contacts in the native state. Interpolating between the two extremes, Shoemaker *et al.* arrived at the following mean field approximation to the contact entropy loss in a partially structured folding ensemble:

$$S_{ij} = +k_B \log[\Delta V/(|i - j|^{-3/2} + (\mu/N)^{-3/2})].$$

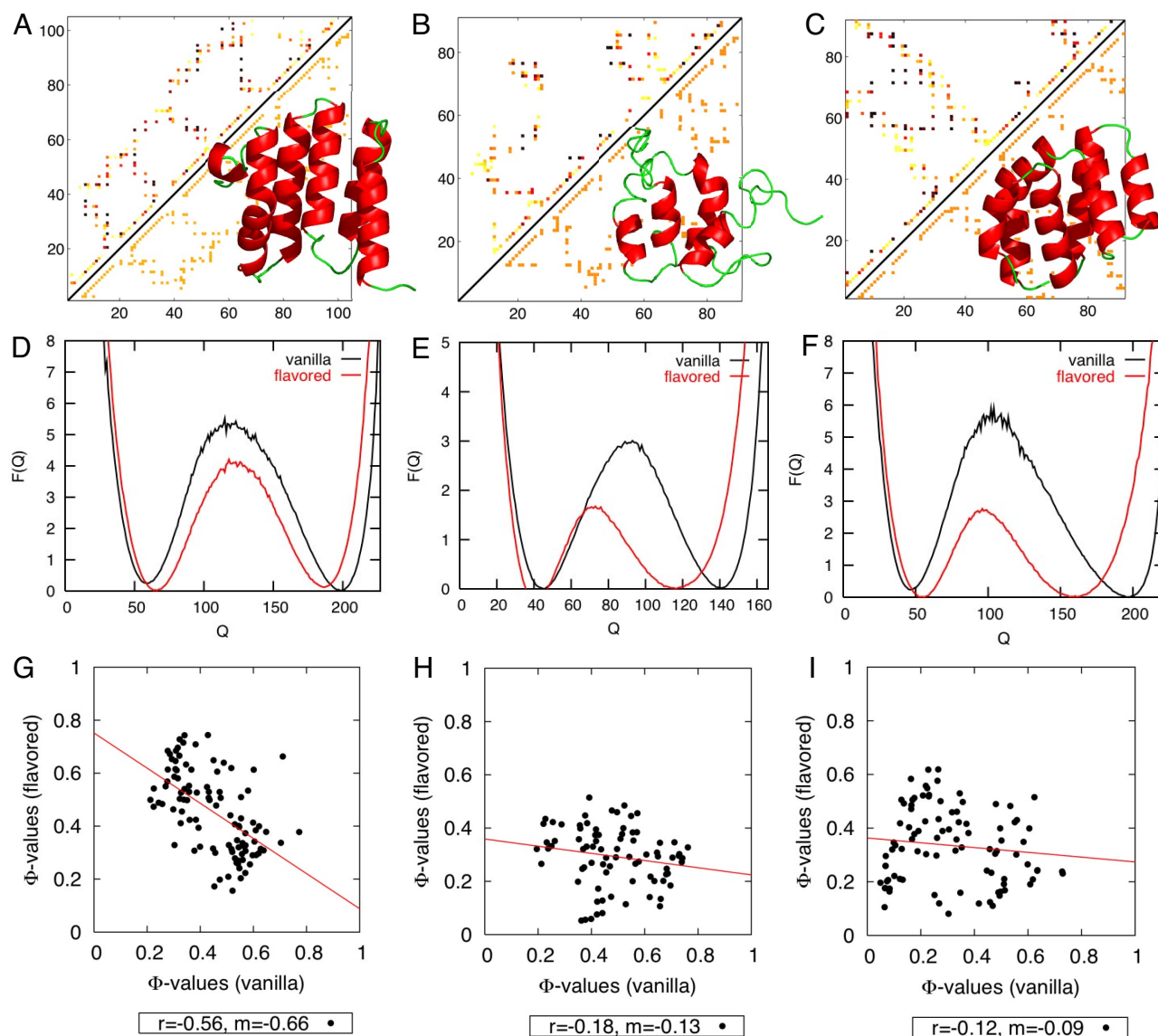


Fig. 3. The folding mechanisms of three all- α proteins (from *Left to Right*, 1v54E0, 1f6vA0, and 1cy5A0) selected from the CATH database. (A–C) The matrices of the interaction energies in the vanilla and flavored models are plotted below and above the diagonal, respectively, with darker colors representing stronger interactions. The corresponding native structures are also shown. (D–F) From simulations of the vanilla and flavored models, the free energy profiles were generated with respect to the order parameter Q . (G–I) The Φ values from the vanilla and flavored models are compared in a plot with a best-fit line.

The resulting free energy functional takes the form:

$$\begin{aligned}
 F(Q_{ij}(\mu)) = & \sum_{ij} \varepsilon_{ij} Q_{ij}(\mu) \\
 & + T \left(\sum_{ij} S_{ij} Q_{ij}(\mu) + \sum_{\mu'=1}^{\mu} \sum_{ij} (\partial S_{ij}(\mu') / \partial \mu') \delta Q_{ij}(\mu') + N \log(\nu) \right) \\
 & + T \left(\sum_{ij} Q_{ij} \log(Q_{ij}(\mu)) + (1 - Q_{ij}(\mu)) \log(1 - Q_{ij}(\mu)) \right)
 \end{aligned}$$

where $\delta Q_{ij}(\mu') = Q_{ij}(\mu') - Q_{ij}(\mu' - 1)$, and the final term accounts for the different ways of forming a contact in a partially ordered protein. The entropy lost as the chain goes from the unfolded to folded states is estimated as $N \log(\nu)$, where ν is the number of conformations per residue. This is essentially the free energy

function of an inhomogeneous-field Ising magnet. The inhomogeneity contains both an entropic and an energetic part.

When the mean-field expressions for the energy and entropy of ensembles from the simulations are stratified with respect to Q , both the entropy and energy, on average, are nearly linearly related to Q (Fig. 4 *A* and *B*) for both proteins. On the other hand, the fluctuations, as quantified by the variance, of the entropy costs of forming contacts $\langle \delta S^2 \rangle$ at Q value and the energies of the formed contacts $\langle \delta \varepsilon^2 \rangle$ show different trends for each protein (Fig. 4 *C* and *D*). By comparing the quantity $\langle \delta S^2 \rangle / \langle \delta \varepsilon^2 \rangle$ at the transition state for each of the proteins, we can quantify which of the two contributions to the “random” fields will dominate the pattern of contacts formed. The ratio determines whether the entropic or energetic fluctuations dominate the folding mechanism. A high (low) value indicates that entropic (energetic) fluctuations determine the structure of the transition state ensemble. It is noteworthy that the ratio $\langle \delta S^2 \rangle / \langle \delta \varepsilon^2 \rangle$ is strongly correlated with the above-mentioned $N_{\text{long}} / N_{\text{short}}$, with a correlation coefficient of 0.90 (Fig. 5). Therefore, for

cannot be achieved in the laboratory, these general trends agree with the arguments based on the free-energy functional of a β -protein with enhanced native contact heterogeneity (26). Also, a marked difference in the Φ values exists (Fig. 6D), as was seen earlier only for the all- β proteins. Therefore, with a sufficiently large $\langle \delta e^2 \rangle$, albeit in an unrealistic regime, the entropy costs intrinsic to forming the topology of the protein are no longer the sole significant factors in folding. The correlation between the Φ values of the vanilla as compared to those of the various flavored models disappears at a lower value of χ in the lambda repressor than the src-SH3 domain (Fig. 6E). In both proteins, the Φ values of the flavored models remain close to that of the vanilla model if $\langle \delta S^2 \rangle / \langle \delta e^2 \rangle$ is greater than around 0.60 (Fig. 6F).

Conclusions

Near the end of the movie “Magnum Force” (1973), Dirty Harry famously tells the villain, “A man’s got to know his limitations.” Likewise, a good theoretician must understand the limitations of models and appreciate the regimes where they will fail. Toward that end we hope to have clarified when folding mechanisms can be predicted from simple, homogeneously weighted native structure-based models and when the details of the energetics must be better understood. Indeed, the dispersion of entropy needed to form the transition state is very often dominant. In other cases, these entropy cost fluctuations can be overcome by the fluctuations in energetic reward for forming specific contacts for many proteins, usually having all- α topologies. It seems that the folding mechanisms of all- α proteins generally are more sensitive to energetic heterogeneity than those of more long-range topologies. The details of the folding of all- α proteins cannot always be very accurately predicted from homogeneously weighted native-topology-based models because the strength of the individual interactions can dramatically change the folding mechanism. Nevertheless, even when the details

are hard to predict, the large differences in the folding mechanisms often found for all- β structural homologues of nearly identical structures can be fully understood within the framework of the energy landscape theory. Because of the sensitivity of the folding mechanism to energetic heterogeneity, the detailed mechanistic predictions are not trivial for these systems. Although strong nonnative contacts slow the kinetics of proteins because of friction, the presence of weak nonnative interactions has been shown analytically to increase the folding rate (38) by reducing the entropy of the unfolded state by collapse, and such interactions may generally play a significantly greater role in all- α as compared to other proteins (39). The same may be true for nonadditive interactions that arise from the presence of water and side chains that are absent in our models but have been shown to be important in determining the transition state ensemble for the last assembly events (12). Regardless, energy landscape theory explains why, in very many cases, folding is not as difficult to understand as some still fear (40) and even gives us a quantitative understanding of the limitations of the simplest versions of the folding funnel.

Materials and Methods

In our study, we used a C_α native-topology-based model where a single bead centered on the C_α position represents a residue, as described previously (6) with homogeneous native contact energies (“vanilla model”). The set of energetic weights of the Miyazawa–Jernigan potential (30) was the basis for introducing native energetic heterogeneity to the model (“flavored model”). A detailed description of both models is presented in *SI Materials and Methods*.

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Supporting Information

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SI Materials and Methods

Native Topology-Based Model. In our study, we used a C_α native topology-based model where a single bead centered on the C_α position represents a residue, as described previously (1), with minor changes to introduce native energetic heterogeneity. In this model, the bond and angle potentials string together the beads to their neighbors along the protein chain. The dihedral potential encodes the secondary structure. The protein's native topology defines the network of favorable long-range tertiary interactions, whereas all other nonbonded interactions are repulsive. The energy function for a C_α native topology-based model with configuration Γ is as follows:

$$H(\Gamma, \Gamma_0) = H_{\text{backbone}} + H_{\text{nonbonded}}$$

$$H_{\text{backbone}} = \sum_{\text{bonds}} K_r (r - r_0)^2 + \sum_{\text{angles}} K_\theta (\theta - \theta_0)^2$$

$$+ \sum_{\text{dihedrals}} K_\phi^{(n)} [1 - \cos(n(\phi - \phi_0))]]$$

$$H_{\text{nonbonded}} = \sum_{i < j - 3}^{\text{native}} \varepsilon_1(i, j) \left[5 \left(\frac{\sigma_{i,j}^{\text{nat}}}{r_{i,j}} \right)^{12} - 6 \left(\frac{\sigma_{i,j}^{\text{nat}}}{r_{i,j}} \right)^{10} \right]$$

$$+ \sum_{i < j - 3}^{\text{nonnative}} \varepsilon_2(i, j) \left(\frac{\sigma_{i,j}^{\text{non}}}{r_{i,j}} \right)^{12}.$$

The K_r , K_θ , and K_ϕ are the force constants of the bonds, angles, and dihedral angles, respectively. The r , θ , and ϕ are the bond lengths, the angles, and the dihedral angles, with a subscript zero representing the corresponding values taken from the native configuration, Γ_0 . The nonbonded contact interactions, $H_{\text{nonbonded}}$, contain Lennard-Jones 10–12 terms for the nonlocal native interactions and a short-range steric repulsive term for the nonnative pairs, corresponding to a perfectly funneled energy landscape. We chose as parameters of the energy function $K_r = 100\varepsilon$, $K_\theta = 20\varepsilon$, $K_\phi^{(1)} = 1.0\varepsilon$, and $K_\phi^{(3)} = 0.5\varepsilon$. The energetic weights are defined as $\varepsilon_1 = \varepsilon_2 = \varepsilon$ in the homogeneously weighted native-topology-based model. Energetic heterogeneity is introduced by having ε_1 equal the value of the corresponding weight according to the set of energetic weights of the Miyazawa–Jernigan potential (2), divided by the average value such that the resulting set's average is equal to ε . $\sigma_{i,j}^{\text{nat}}$ is the distance between the C_α atoms of the residues (i, j) in the native configuration and $\sigma^{\text{non}} = 4.0 \text{ \AA}$ for all nonnative residue pairs. The network of native contact pairs was determined by using the CSU (Contacts of Structural Units) software (3). Multiple trajectories

with numerous unfolding/folding transitions were collected and analyzed by using the weighted histogram analysis method (WHAM) (4) to calculate the free energy surface projected onto the fraction of native contacts (Q). This reaction coordinate was previously demonstrated to accurately map to P_{fold} at the resolution of Φ values for a funneled energy landscapes (5). Once the free energy profile has been determined using Q as a reaction coordinate, the structure of the transition state can be described by Φ value analysis. Experimentally, Φ values are determined by a perturbation of the protein sequence by introducing a mutation that effectively deletes the native contacts of the residue. Then, the ratio of the apparent free energy change of the transition state and the folded state with respect to the unfolded state is calculated, resulting in a Φ value for the residue of interest: $\Phi_i^{\text{exp}} = \Delta\Delta G_i^{\text{TS-U}} / \Delta\Delta G_i^{\text{F-U}}$ (6). The equivalent measure from simulations is to compute the $\Phi_{i,j}$ value for each native contact pair between residues i and j from the probability of formation $P_{i,j}$:

$$\Phi_{i,j}^{\text{sim}} = \frac{\Delta\Delta G^{\text{TS-U}}}{\Delta\Delta G^{\text{F-U}}} \approx \frac{P_{i,j}^{\text{TS}} - P_{i,j}^{\text{U}}}{P_{i,j}^{\text{F}} - P_{i,j}^{\text{U}}}.$$

The Φ_i value of residue i , which can be directly compared to experimentally observed values, can be calculated by averaging over $\Phi_{i,j}$ that are involved in residue i : $\Phi_i^{\text{sim}} = (1/n) \sum_j^n \Phi_{i,j}^{\text{sim}}$. The folding temperature (T_f) was identified as the peak of a specific heat versus temperature profile.

In the case of the lambda repressor, we also performed simulations with the “flavored” model where the short-range interactions were reverted to the corresponding vanilla values. The short-range interactions were defined as any native contacts whose distance in sequence $\Sigma i - j \Sigma \leq 6$ and all others were considered long-range interactions.

Survey of the Nonredundant Set of the Protein Data Bank (PDB). A survey was performed of the all α -helical, α/β , and all β -sheet proteins in a nonredundant set of the PDB (7). Using the DSSP program (8), all α -helical proteins were defined as having at least 70% of their residues in an α -helical conformation and no residues in a β -sheet conformation, α/β proteins as having at least 25% of their residues in an α -helical conformation and at least 25% of their residues in a β -sheet conformation, and all β -sheet proteins as having at least 50% of their residues in a β -sheet conformation and no residues in an α -helical conformation. These criteria yielded a total of 342 protein structures for analysis.

To calculate the ratio between long- and short-range native interactions, short-range interactions were defined as any native contacts whose distance in sequence $i - j \leq 6$ and all others were considered long-range interactions.

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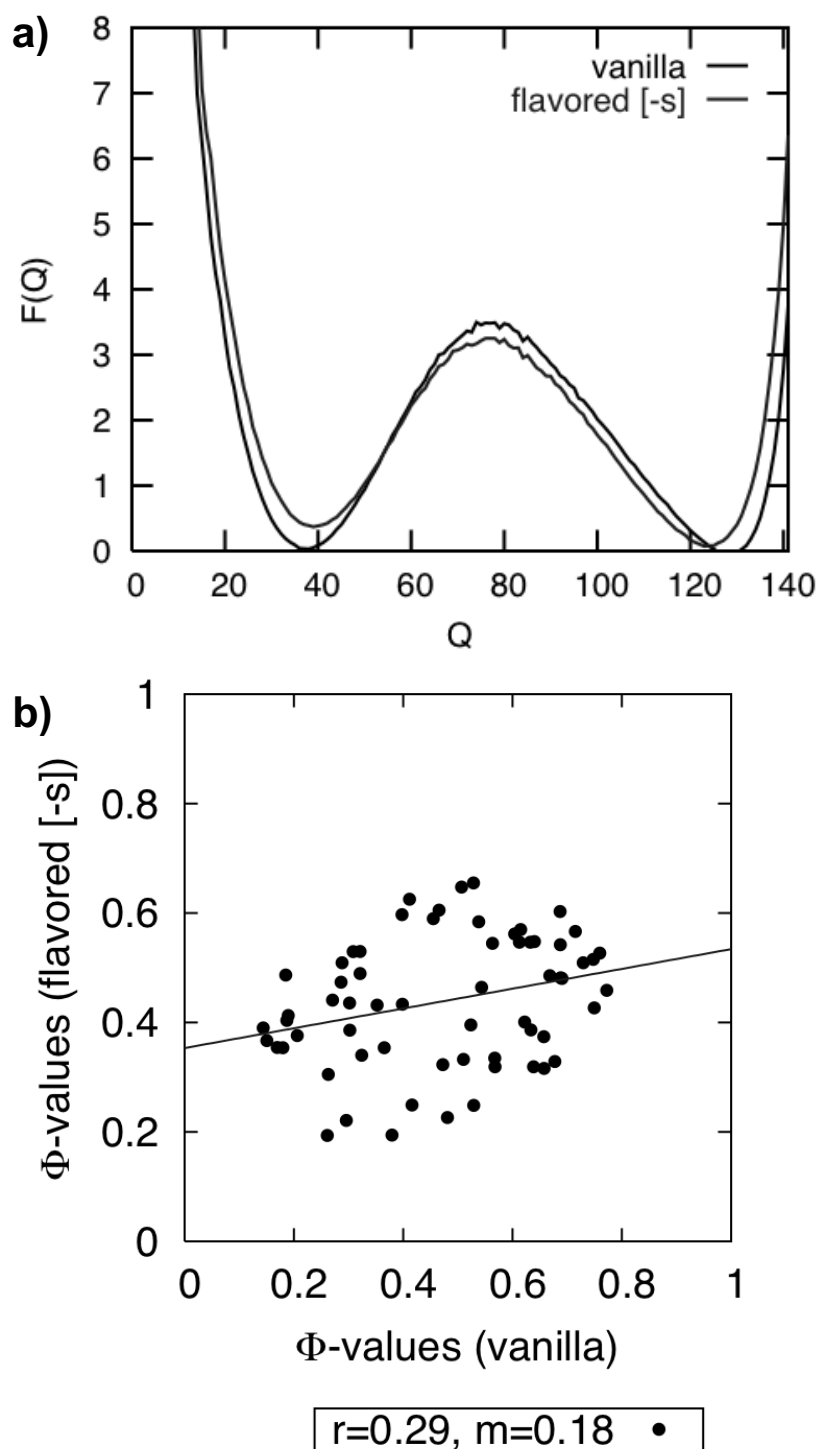
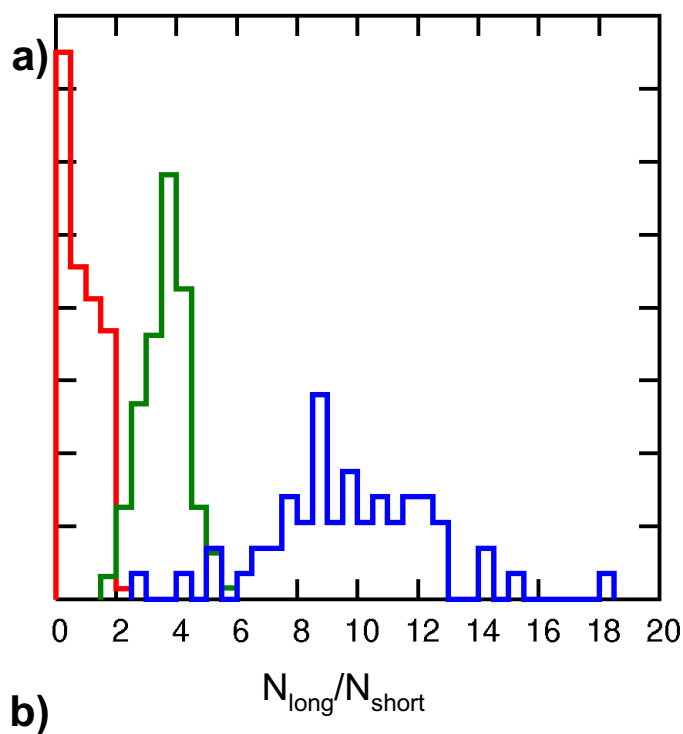


Fig. S1. The flavored model simulation of the lambda repressor, an all- α protein, with the short-range interaction set at the vanilla interaction energies. (A) From simulations, the free energy profiles were generated with respect to the order parameter Q . (B) The Φ values from the vanilla and flavored models with vanilla short-range interaction weights are compared in a plot with a best-fit line.



PDB ID	NumRes	Class	$N_{\text{long}}/N_{\text{short}}$
1R69	63	α	1.45
1IMQ	86	α	1.68
3GB1	56	α/β	3.42
2PTL	60	α/β	3.89
1URN	97	α/β	4.40
2CI2	65	α/β	4.67
1RIS	96	α/β	5.12
1SRL	56	β	7.73

Fig. S2. A comparison of the ratio between long- and short-range native interactions ($N_{\text{long}}/N_{\text{short}}$), in sequence, across the different secondary structural classes. (A) A histogram of the long- and short-range interactions from a survey of the nonredundant set of the PDB. (B) A table of well-studied two-state folding proteins with different number of residues, secondary structure topology, and number of long- and short-range native interactions.